Final Project: Scenario

Estimated reading time: 15 minutes

Introduction

This project focuses on refreshing your skills and understanding of the components of BI and its process. After gathering, cleaning, and transforming the given data, the next step is visualizing it based on the business requirements. Finally, the insights drawn are presented to the stakeholders through compelling visual reports and dashboards.

Learning Objectives

- Evaluate and apply BI techniques, tools, and methodologies to real-world scenarios and business problems.
- Apply the concepts and techniques to analyze and visualize sales data.

Scenario

Imagine you work as a BI analyst at a retail store that sells office supplies in the United States.

Your task is to analyze how well the retail store sells furniture, office supplies, and electronics across the various states in the US. The management team would like some insights from their data about which products and market segments are profitable and where improvements should be focused. The management team wants to know which products bring in the maximum sales revenue, the products that require more focus, and where the revenue is not growing as expected.

Project Overview

The data set sample below is a sample of their online sales analysis for the different market segments and product categories.

A	B	C	D	E	F	G	H	J	K	M	N	O
Row ID	Customer I	Customer Name	Market Segr	State	Product ID	Product Category	Sub-Category	Order ID	Order Date	Sales	Quantit	Profit
1	CG-12520	Claire Smith	Consumer	Ohio	FUR-BO-10001798	Furniture	Bookshelves	SI-2020-152156	11/8/2016	261.96	2	41.91
2	CG-12520	Claire Smith	Consumer	Ohio	FUR-CH-10000454	Furniture	Chairs	SI-2020-152156	11/8/2016	731.94	3	219.58
3	DV-13045	Darrin Van Huff	Corporate	Texas	OFF-LA-10000240	Office Supplies	Labels	SI-2020-138688	6/12/2016	14.62	2	6.87
4	SO-20335	Sean Meyer	Consumer	Arkansas	FUR-TA-10000577	Furniture	Desks	TY-2021-108966	10/11/2015	957.5775	5	254.50
5	SO-20335	Sean Meyer	Consumer	Arkansas	OFF-ST-10000760	Office Supplies	Storage	TY-2021-108966	10/11/2015	22.368	2	2.52
6	BH-11710	SarahAlderman	Consumer	Texas	FUR-FU-10001487	Furniture	Decor	CA-2014-115812	6/9/2014	48.86	7	14.17
7	BH-11710	SarahAlderman	Consumer	Texas	OFF-AR-10002833	Office Supplies	Art	CA-2014-115812	6/9/2014	7.28	4	1.97
8	BH-11710	SarahAlderman	Consumer	Texas	TEC-PH-10002275	Electronics	Phones	CA-2014-115812	6/9/2014	907.152	6	90.72
9	AA-10480	Andrew Allen	Consumer	North Carolina	OFF-PA-10002365	Office Supplies	Paper	AB-2019-114412	4/15/2017	18.504	3	5.78
10	IM-15070	Irene Maddox	Consumer	Washington	OFF-BI-10003656	Office Supplies	Binders	SI-2020-161389	12/5/2016	114.9	5	34.47
11	HP-14815	Sam Pawlan	Small Business	Texas	OFF-AP-10002311	Office Supplies	Appliances	TY-2021-118983	11/22/2015	1706.184	9	85.31
12	HP-14815	Sam Pawlan	Small Business	Texas	OFF-BI-10000756	Office Supplies	Binders	TY-2021-118983	11/22/2015	911.424	4	68.36
13	PK-19075	Pete Kriz	Consumer	Wisconsin	OFF-ST-10004186	Office Supplies	Storage	CA-2014-105893	11/11/2014	15.552	3	5.44
14	AG-10270	Alejandro Grove	Consumer	Utah	OFF-ST-10000107	Office Supplies	Storage	CA-2014-167164	5/13/2014	407.976	3	132.59
15	ZD-21925	Alexis Bauer	Consumer	Texas	OFF-AR-10003056	Office Supplies	Art	CA-2014-143336	8/27/2014	68.81	5	132.20
16	ZD-21925	Alexis Bauer	Consumer	Texas	TEC-PH-10001949	Electronics	Phones	CA-2014-143336	8/27/2014	2.544	3	4.78
17	ZD-21925	Alexis Bauer	Consumer	Texas	OFF-BI-10002215	Office Supplies	Binders	CA-2014-143336	8/27/2014	665.88	6	13.32
18	KB-16585	Ken Moore	Corporate	Nebraska	OFF-AR-10000246	Office Supplies	Art	SI-2020-137330	12/9/2016	55.5	2	9.99
19	KB-16585	Ken Moore	Corporate	Nebraska	OFF-AP-10001492	Office Supplies	Appliances	SI-2020-137330	12/9/2016	8.56	2	2.48
20	SF-20065	Sandra Flanagan	Consumer	North Carolina	FUR-CH-10002774	Furniture	Chairs	US-2017-156909	7/16/2017	213.48	3	16.01
21	EB-13870	Emily Burns	Consumer	Utah	FUR-TA-10000577	Furniture	Desks	CA-2015-106320	9/25/2015	22.72	4	7.38
22	EH-13945	Eric Aldermann	Consumer	Texas	OFF-BI-10001634	Office Supplies	Binders	SI-2020-121755	1/16/2016	19.46	7	5.06
23	EH-13945	Eric Aldermann	Consumer	Texas	TEC-AC-10003027	Electronics	Accessories	SI-2020-121755	1/16/2016	60.34	7	15.69
24	TB-21520	Anne Blumstein	Consumer	North Carolina	FUR-BO-10004834	Furniture	Bookshelves	TY-2021-150630	9/17/2015	71.372	2	5.60
25	TB-21520	Anne Blumstein	Consumer	North Carolina	OFF-AR-10001683	Office Supplies	Art	TY-2021-150630	9/17/2015	1044.63	3	240.26
26	MA-17560	Matt Abelman	Small Business	Texas	OFF-PA-10000249	Office Supplies	Paper	AB-2019-107727	10/19/2017	11.648	2	4.22

Steps in the BI process:

Before you can analyze the data for patterns and insights, you must follow the BI process steps.

1. Gather all relevant data.
2. Clean the data.
3. Perform data mining for insights.
4. Apply BI techniques such as visualization.
5. Communicate your findings.

Step 1: Gather all relevant data

In this example, there is data on customers like their ID, name, state, and market segment. Then, there is information on products like product ID, category, and sub-category. Order data like order ID, order date, sales, quantity, and profit are also provided. The information in three files has been combined into one Excel spreadsheet for easy analysis.

Step 2: Clean the data

Data needs to be cleaned and normalized where required. Check the date columns to see if the format is right. Make sure the quantities are whole numbers. Check for any nulls or empty rows, and see if you can return to the data sources and get the missing data. Remove duplicates, if any.

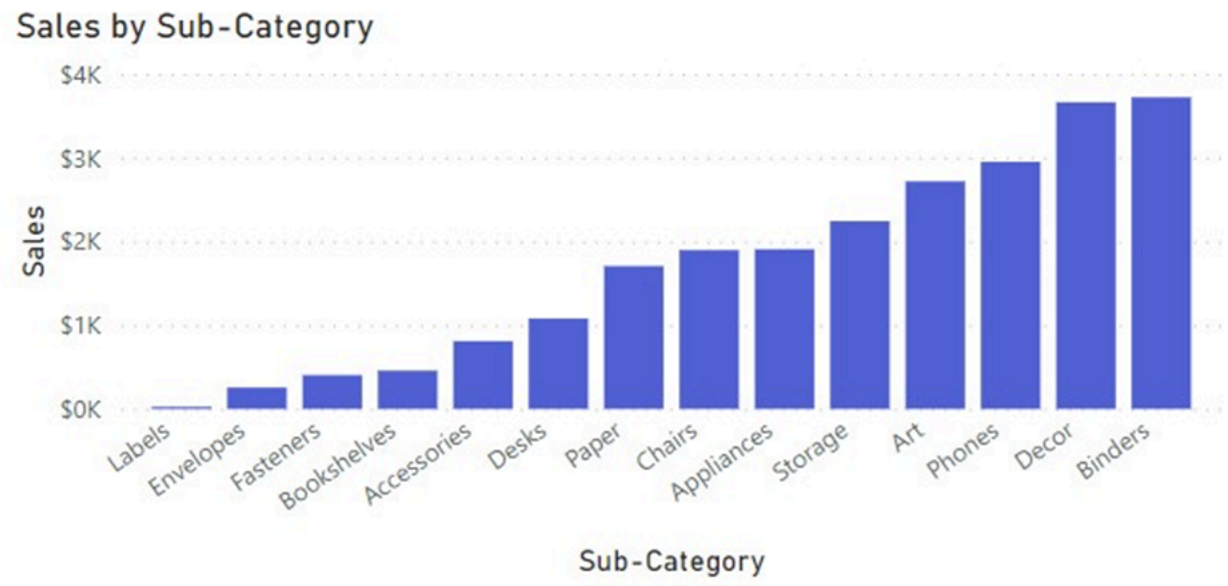
Step 3: Mining the data for insights

Use spreadsheets for performing basic data mining tasks. Excel add-ins for data mining allow for more extensive tasks such as classification, regression, association rules, clustering, and model building.

Step 4: Visualization

Use a BI tool such as Microsoft Power BI or Excel to generate charts and graphs to visualize data to share your insights.

Step 5: Communicate your findings

Use tools such as IBM Cognos Analytics, Google Looker, or Tableau to create compelling reports and dashboards to communicate your findings and insights about the data with the relevant stakeholders.

The graph below samples one such visualization that you would use to capture a trend hidden in the sample data set.